TECHNICAL PORTFOLIO

2026

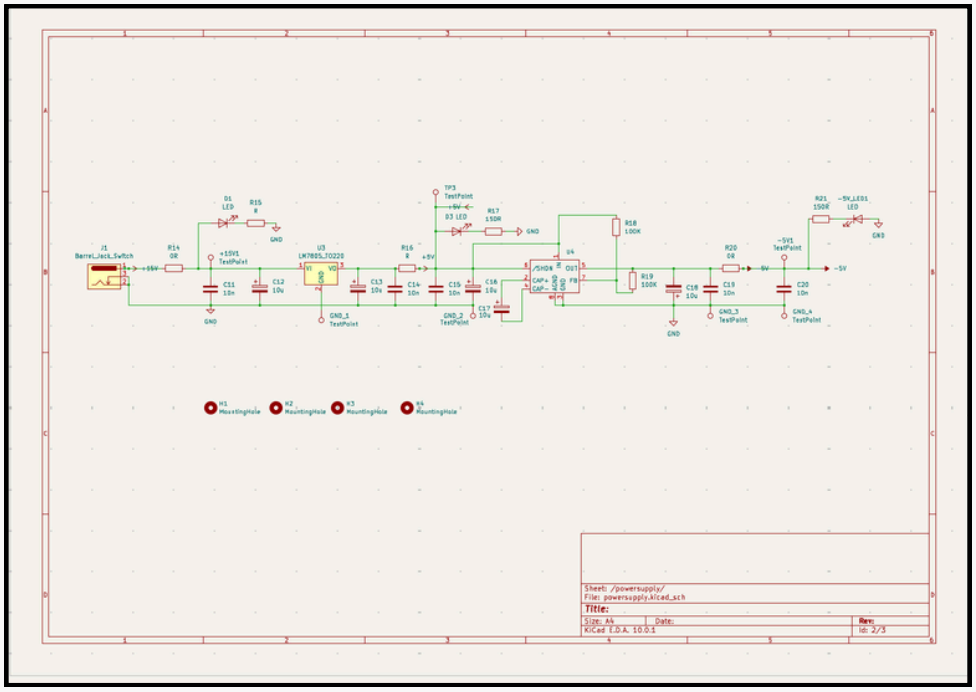
Sophia Liau

4th year biomedical engineering student @ UBC.

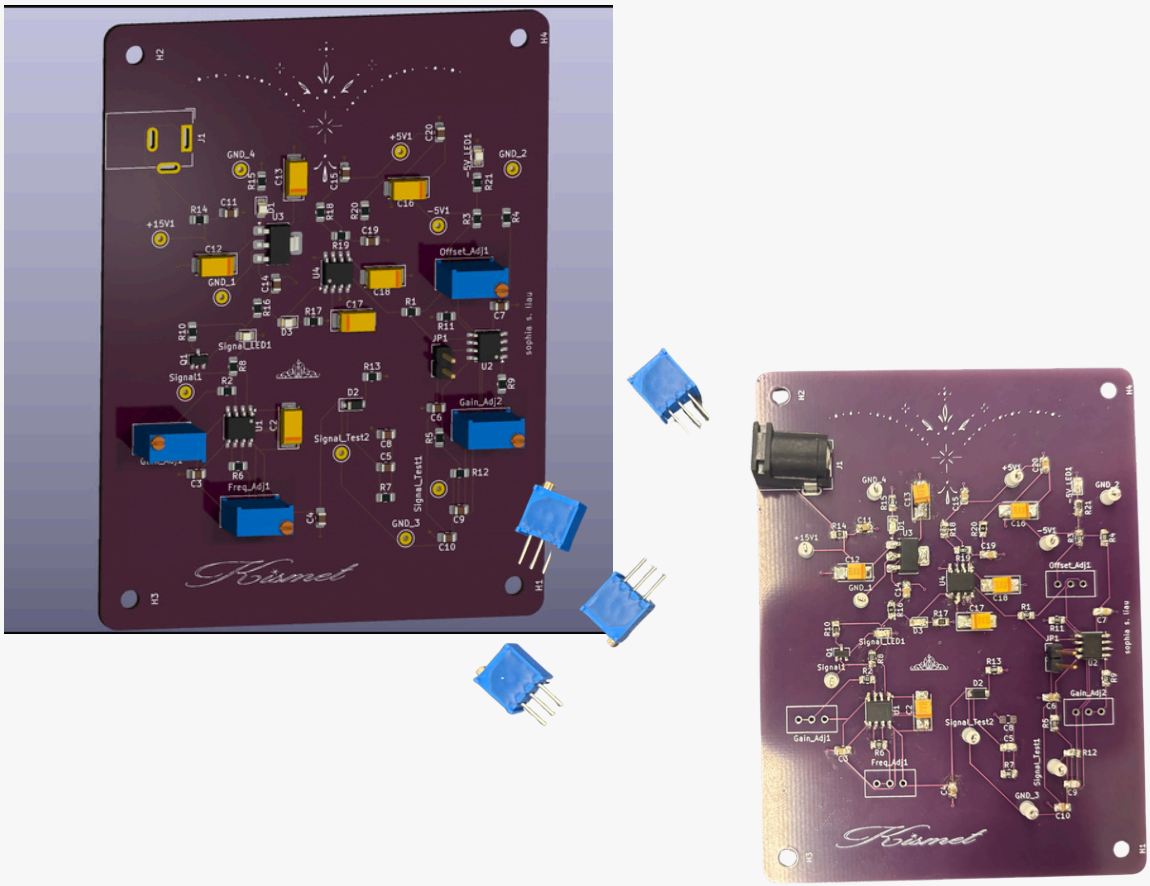
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PCB DESIGN

2026

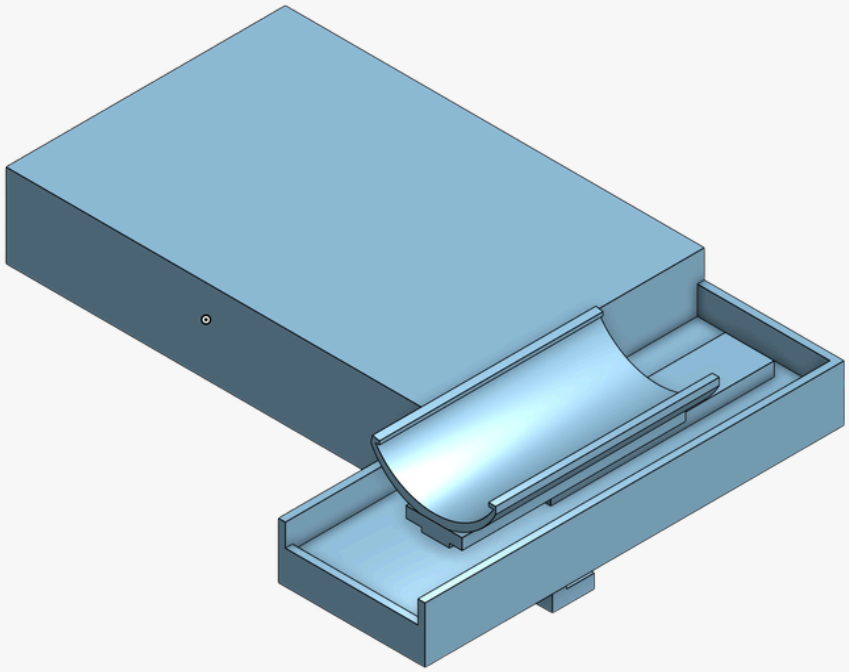


Designed a flashing-light circuit schematic in KiCad and assembled and soldered the complete PCB. Performed software-based checks and tested the circuit throughout soldering and assembly to identify issues as the build progressed.

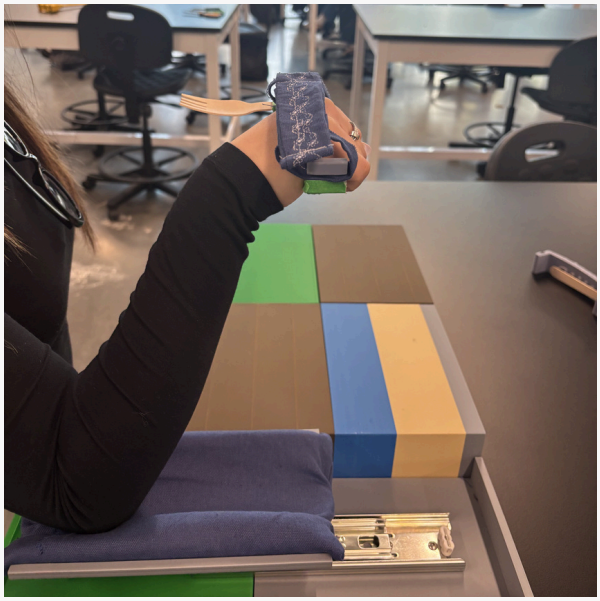


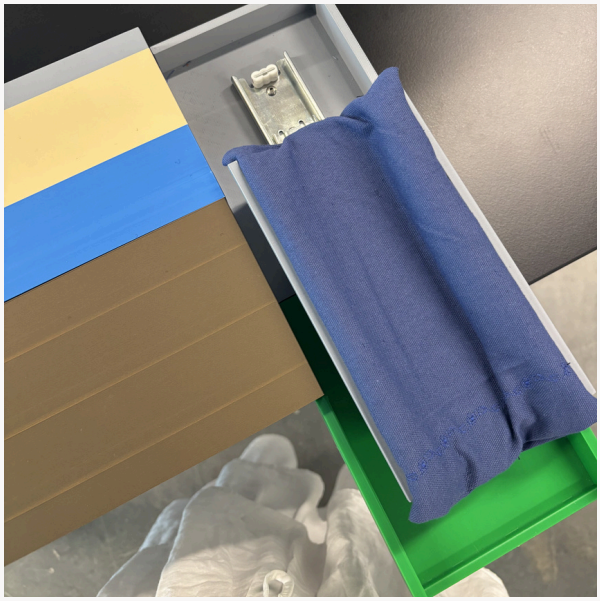
GLIDEFLEX SYSTEM

2026



Designed and built an assistive feeding device for children with cerebral palsy. My contributions included modelling components in Onshape, 3D printing and assembling parts, and sewing the supporting pillow and strap. The project combined mechanical design and hands-on fabrication to address accessibility and user comfort.



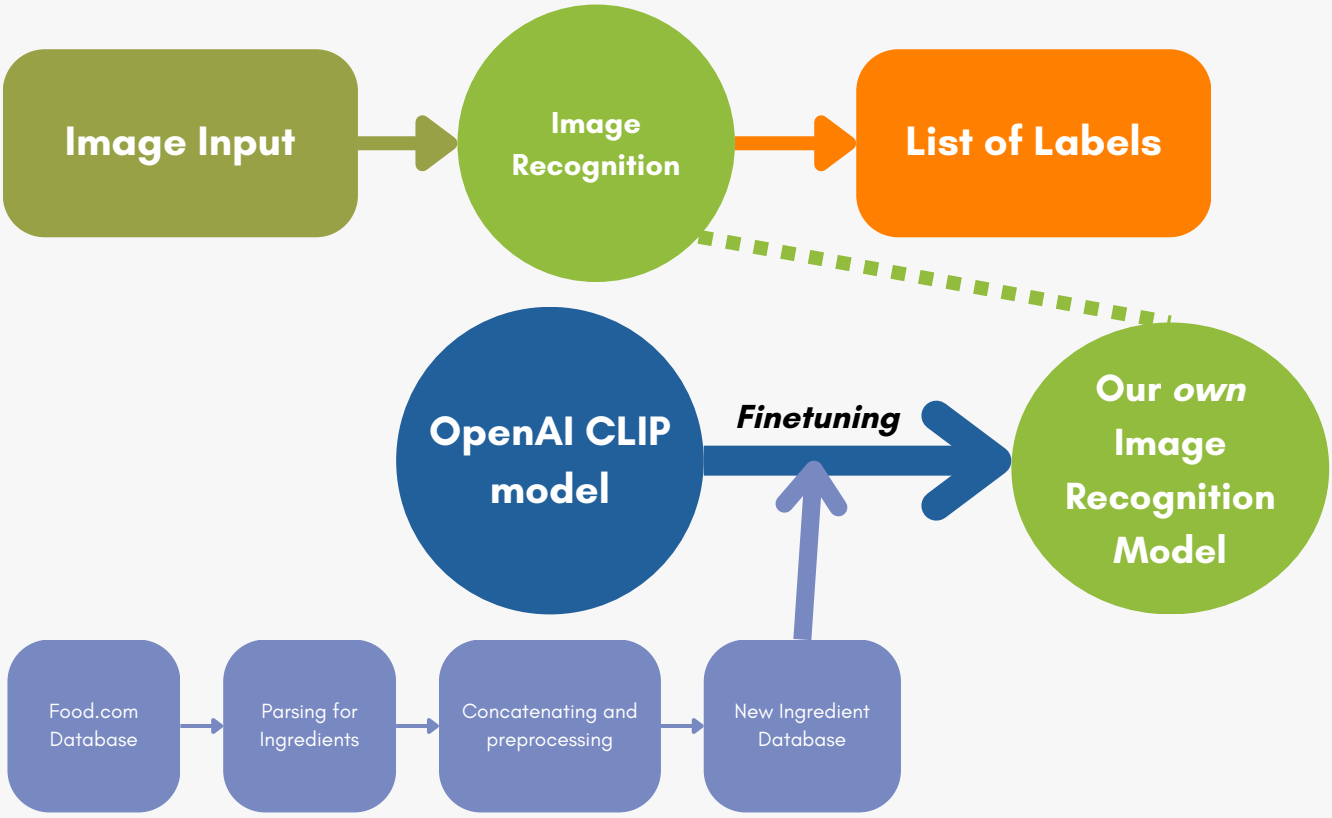


SUSTAINABITE

2025



Sustainabite is a prototype recipe-recommendation application that combines image recognition and machine learning to suggest recipes based on available ingredients. My contribution focused on developing two additional models to predict cuisine category and estimate cooking time, helping tailor recommendations to users' preferences.



BUZZY BUDDY BAND

2025



The Buzzy Buddy Band is a wearable device designed to assist diabetic children with medication adherence. It utilizes a combination of sensors and haptic feedback to alert users to actionable changes in blood glucose, all in a child-focused watch design. I primarily worked on mechanical components with a team to create an innovative solution that enhances accessibility and safety for our target users and their families.



